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Utility vehicle tyre chain having at least one cleat plate, utility vehicle tyre and cleat plate for a utility vehicle tyre chain

Abstract

A utility vehicle tire chain, in particular for forestry vehicles, tractors, or wheel loaders, has a plurality of cleat plates directly or indirectly connected to one another via chain strands, wherein each cleat plate has at least one cleat to improve traction which extends away from an upper side of the cleat plate. The solution according to the invention can simplify the manufacture of a utility vehicle tire chain and improve its stability.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS

(1) The present application is a national stage entry under 35 U.S.C. § 371 of International Application No. PCT/EP2020/050311, filed on Jan. 8, 2022, which claims the benefit of German Patent Application No. 10 2019 200 350.8, filed on Jan. 14, 2019. The entire content of the aforementioned applications is hereby incorporated by reference.

BACKGROUND OF THE INVENTION

(2) The invention relates to a utility vehicle tire chain, in particular for forestry vehicles, tractors and wheel loaders. The invention additionally relates to a utility vehicle tire and a cleat plate for a utility vehicle tire chain.

(3) In the field of forestry utility vehicles, tire chains having large rings that rest on the tire tread areas are currently used to increase traction on slippery surfaces. These rings are made from relatively strong wire having a diameter of 19 mm to over 30 mm. Due to the increased requirements with regard to the soil compaction produced by forestry utility vehicles, larger tires are increasingly being used. A larger ring has previously simply been used for larger tires. However, as the size increases, these rings are more difficult to manufacture. The heat treatment becomes problematic because of the large diameter. In addition, high forces are required for bending.

(4) The invention is therefore based on the object of creating a utility vehicle tire chain, in particular for forestry vehicles, which is easy to manufacture despite increasingly larger tire sizes and, in particular, is also easy to scale with the tire size.

BRIEF SUMMARY OF THE INVENTION

(5) This object is satisfied according to the invention by a utility vehicle tire chain, in particular for forestry vehicles, tractors, or wheel loaders, which comprises a plurality of cleat plates directly or indirectly connected to one another via chain strands, wherein each cleat plate comprises at least one cleat to improve traction which extends away from an upper side of the cleat plate. For the utility vehicle tire mentioned at the outset, the object according to the invention is satisfied in that the utility vehicle tire is provided with a utility vehicle tire chain according to the invention.

(6) The above-mentioned problems of known utility vehicle tire chains can be solved by the solution according to the invention. Cleat plates are easy to manufacture. In addition, cleat plates have a good load-bearing capacity, in particular compressive strength.

(7) The cleat plate can comprise in particular a main body which is substantially plate-shaped. In other words, the main body is formed by a plate which is flat and spans a plate plane. In particular, the main body, preferably the entire cleat plate, can be made from metal, in particular from sheet metal. The upper side of the cleat plate is the side which, when the utility vehicle tire chain is mounted, extends away from the utility vehicle tire and comes into contact with the ground. The upper side is provided with the at least one cleat. When the utility vehicle tire chain is mounted, an underside of the cleat plate opposite the upper side can rest on a tread surface of the utility vehicle tire. The underside can have a base area of the cleat plate or be identical thereto.

(8) The solution according to the invention can be further improved by various configurations that are advantageous on their own and can be combined at random. These configurations and the advantages associated therewith shall be described below.

(9) According to a first advantageous embodiment of the utility vehicle tire chain, at least one cleat can be welded to a plate-shaped main body of the cleat plate. A cleat can be quickly and reliably connected to the main body by welding. As an alternative or in addition thereto, at least one cleat can be screwed to the main body or attached to the main body in some other way. As an alternative to the separate manufacture of the main body and the at least one cleat with the subsequent connection of the parts, the at least one cleat can be manufactured monolithically with the main body, in particular by forging.

(10) At least one of the cleat plates can have a rectangular, in particular a square, main body. As an alternative thereto, the main body can also be round, oval, or polygonal. The shape there does not relate to a cross-section through the plate, but rather to the two-dimensional extension of the plate which is visible when looking onto the upper side or the lower side. In other words, a base surface resting on a tire tread area in the mounted state has the respective shape.

(11) The rectangular, in particular square shape of the main body does not rule out rounded corners. It is instead essential that the main body comprise four sides which are arranged with respect to one another substantially like the sides of a rectangle.

(12) In order to reduce the weight of a cleat plate and/or to save material during production, at least one of the cleat plates can have a plate-shaped main body with at least one recess extending into the main body transverse to a plate plane.

(13) According to a preferred embodiment, the recess extends through the main body. In other words, the main body has an opening or a hole that penetrates the main body. The at least one recess is preferably circular. If the cleat plate comprises a plurality of cleats, then the at least one recess is preferably arranged between these cleats, preferably at the center of the plate.

(14) In order to connect at least one of the cleat plates in a simple and reliable manner to the chain strands, at least one of the cleat plates has a plate-shaped main body with at least two receptacles, in each of which an end link of a chain strand is received. The end links of the chain strands are preferably welded to the main body of the cleat plate. The receptacles can be formed as recesses in the plate through which the main body is open towards its sides.

(15) As an alternative or in addition to the attachment of end links of the chain strands in the receptacles, end links can also be connected directly to the cleat plate. For this purpose, at least one end link can be welded or otherwise connected to the cleat plate, in particular to a main body of the cleat plate. In order to simplify the manufacturing process, the cleat plate, in particular its base plate, can be provided with at least one recess into which an end link can be placed. As a result, the position of the end link can be specified and the end link can be secured against slippage during the connection.

(16) It is not necessary to connect an end link of a chain strand to a cleat plate. If at least one chain strand comprises an element which is no chain link at its end to be connected to the cleat plate, then the chain strand can be connected to the cleat plate via this element. For example, the chain strand can be provided with a ring or a shackle at one end. In this case, the ring or the shackle can then be connected to the cleat plate. Each of the aforementioned variants can be used to attach a chain strand to a cleat plate.

(17) A cleat height, measured in particular from the upper side of the main body to the end of the cleat, preferably corresponds to at least twice the thickness of a plate-shaped main body of the cleat plate. The height of the cleat is preferably at least 40 mm, particularly preferably 50 mm to 70 mm. A thickness of the cleat preferably corresponds to the plate thickness. At least one cleat and the at least one main body can be made from the same original material. A length of the cleat is preferably at least 60 mm, particularly preferably at least 80 mm.

(18) At least one cleat can be provided with a tip extending away from the remainder of the cleat in

order to further improve traction. At least one cleat is preferably aligned parallel to one side of a rectangular main body. Particularly preferably, at least one cleat plate is provided with four cleats, wherein each of the cleats is aligned parallel to one of the sides of the rectangular shape. The cleats are there preferably spaced equidistantly from a center of the cleat plate. A recess extending preferably entirely through the main body can be arranged between the four cleats. In a top view onto the cleat plate, the cleats can have an elongate shape. Alternatively, they can also have a zigzag shape, a triangular shape, an arrow shape, or some other shape.

(19) In order to obtain a stable cleat plate, at least one of the cleat plates can comprise a plate-shaped main body, the thickness of which is at least 15 mm, preferably 20 to 30 mm. A total height of a cleat plate, which is composed of the thickness of the main body and the height of the cleat, is preferably at least 70 mm, particularly preferably at least 90 mm. The main body is preferably formed from sheet metal. At least one cleat as well.

(20) In order to further increase the traction, the utility vehicle tire chain can comprise at least two rows of cleat plates extending along a circumferential direction of the utility vehicle tire chain. At least two cleat plates can be arranged adjacent to one another, in other words at the same height, transverse to the circumferential direction. The at least two cleat plates can be in particular cross-connected to one another. Alternatively, a utility vehicle tire chain according to the invention can also comprise one row or more than two rows of cleat plates. If at least two rows are present, then cleat plates of the two rows can be offset from one another in the circumferential direction instead of being arranged at one height. If there are more than two rows, combinations are also possible. This means, in the case of, for example, three rows, the cleat plates of two rows can be arranged at one height in the circumferential direction. The cleat plates of the third row can be offset in the circumferential direction relative to the other cleat plates.

(21) Each of the cleat plates can have four sides, wherein each cleat plate is connected on three sides via chain strands to at least one adjacent cleat plate and on one side to at least one side chain or one side strand, respectively.

(22) According to one advantageous embodiment, the utility vehicle tire chain is symmetrical with respect to a plane of symmetry extending transverse to a tire axis. In an unmounted state, this plane of symmetry is perpendicular to a plane of the mesh formed from the utility vehicle tire chain and parallel to a longitudinal direction or circumferential direction of the utility vehicle tire chain. Due to the symmetrical configuration, the utility vehicle tire chain can be independent of the direction of travel. In other words, it is not necessary during assembly to ensure that a predetermined direction of travel is adhered to.

(23) In order to reduce or avoid bending of a cleat plate under load, at least one cleat plate is preferably smaller than a tire tread area. The size of the tire tread area results from the size of the tire for which the utility vehicle tire chain is intended. It is to be noted here that each utility vehicle tire chain is generally manufactured for a predetermined type of vehicle tire. The types of vehicle tires are standardized and known to the person skilled in the art.

(24) At least one cleat can be arranged such that it bridges a receptacle for an end link or another element of a chain strand. The longitudinal side of the cleat can extend transverse to the direction along which the associated receptacle extends into the material of the main body. This direction is preferably the same direction in which the depth of the receptacle extends. The cleat thickness can substantially correspond to the depth of the receptacle. Due to the cleat bridging a receptacle, a compact structure of a cleat plate is made possible. In addition, the at least one cleat can form a protection for the receptacle that it bridges. Likewise, the at least one cleat can protect an end link received in the receptacle or another part of a chain strand received in the receptacle.

(25) At least one cleat, which is arranged such that it bridges a receptacle, can respectively comprise on its underside a recess extending into the cleat. The underside of the cleat is the side that faces the main body. The recess has a length which preferably corresponds to at least the width of the receptacle for an end link. For the sake of simplicity, the length of the recess can be equal to

the length of a tip of the cleat. In the fully mounted state, an end link of a chain strand can protrude into the recess. A height of the recess is preferably selected such that an end link arranged in the receptacle does not hit the cleat.

(26) The invention shall be explained hereafter in more detail by way of example using advantageous embodiments with reference to the drawings. The feature combinations illustrated in the embodiments by way of example can respectively be supplemented subject to the above explanations by other features for a particular case of application. Individual features can also be omitted in the embodiment described subject to the above explanations if the effect of this feature is irrelevant in a specific case of application.

Description

BRIEF DESCRIPTION OF THE SEVERAL VIEWS OF THE DRAWINGS

(1) The same reference numerals in the drawings are always used for elements of the same function and/or the same structure, wherein:

(2) FIG. 1 shows a top view onto a part of a first advantageous embodiment of a utility vehicle tire chain according to the invention;

(3) FIG. 2 shows an enlarged illustration of a cleat plate of the utility vehicle tire chain from FIG. 1;

(4) FIG. 3 shows a side view of the cleat plate from FIG. 2;

(5) FIG. 4 shows a perspective illustration of a cleat plate of the first embodiment without chain strands;

(6) FIG. 5 shows a main body of the first embodiment of the cleat plate;

(7) FIG. 6 shows a side view of a cleat of the first embodiment of the cleat plate;

(8) FIG. 7 shows a second embodiment of a utility vehicle tire chain according to the invention in the unmounted state in a perspective illustration;

(9) FIG. 8 shows the second embodiment of the utility vehicle tire chain in a mounted state on a utility vehicle tire;

(10) FIG. 9 shows a third embodiment of a utility vehicle tire chain according to the invention with a third embodiment of cleat plates in a perspective illustration;

(11) FIG. 10 shows a main body for a third embodiment of a cleat plate according to the invention;

(12) FIG. 11 shows a schematic illustration of a cleat of the third embodiment; and

(13) FIG. 12 shows another embodiment of a utility vehicle tire chain according to the invention with the third embodiment of cleat plates offset from one another.

DETAILED DESCRIPTION OF THE INVENTION

(14) In the following, a first embodiment of the utility vehicle tire chain according to the invention shall be described with reference to FIG. 1. The details of the cleat plates shall thereafter be discussed with reference to FIGS. 2 to 6.

(15) Utility vehicle tire chain 1 comprises a plurality of cleat plates 3. Cleat plates 3 are connected to one another via chain strands 5. Some of chain strands 5 serve as longitudinal connections 7 which run parallel to a circumferential direction U of utility vehicle tire chain 1.

(16) Chain strands 5 arranged transverse to circumferential direction U and between two adjacent cleat plates 3 serve as transverse connections 9 of utility vehicle tire chain 1. Further chain strands 5 running transverse to circumferential direction U serve as side strands 10 of utility vehicle tire chain 1. Cleat plates 3 can be indirectly connected to one another via side strands 10.

(17) Utility vehicle tire chain 1 comprises two rows 11 of cleat plates 3, wherein each row 11 extends parallel to circumferential direction U. Two cleat plates 3 each from two rows 11 are arranged opposite to one another transverse to circumferential direction U. In other words, two cleat plates 3 are arranged at the same height along circumferential direction U.

(18) Utility vehicle tire chain **1** is symmetrical with respect to a plane of symmetry **15** running parallel to circumferential direction **U** and perpendicular to a continuous chain mesh plane **13** spanned by utility vehicle tire chain **1**. In the mounted state, plane of symmetry **15** runs transverse to a tire axis (presently not shown).

(19) Other configurations are also possible as an alternative to the symmetrical configuration of utility vehicle tire chain **1** in which two cleat plates **3** are arranged at the same height along circumferential direction **U**. For example, two rows **11** could be offset from one another along circumferential direction **U**, wherein transverse connections **9** are then able to connect cleat plates **3** to one another in a zigzag arrangement.

(20) Cleat plates **3** are substantially square. This means that they have longitudinal sides **17**, two of which are disposed opposite to one another and run parallel to one another, and wherein two longitudinal sides **17** that are not disposed opposite to one another run perpendicular to one another.

(21) In particular a rectangular shape, a circular shape, or a polygonal shape are possible as an alternative to the square shape. In the first embodiment, cleat plates **3** are aligned within utility vehicle tire chain **1** such that two longitudinal sides **17** each run parallel to circumferential direction **U** and two longitudinal sides **17** run transverse to circumferential direction **U**.

(22) The structure of a cleat plate **3** according to the invention shall be described below with reference to FIGS. **2** to **6**.

(23) Cleat plate **3** comprises a total of eight receptacles **19** for end links **21** of chain strands **5**. Receptacles **19** are in particular slot-shaped.

(24) End links **21** of chain strands **5** can be inserted into receptacles **19** and welded to cleat plate **3**. Consequently, end links **21** are connected by way of welded connections **23** to regions of cleat plate **3** surrounding receptacles **19**. Receptacles **19** preferably extend straight from longitudinal sides **17** into the material of cleat plate **3**. Each receptacle **19** preferably runs transverse to longitudinal side **17** on which it is arranged.

(25) Receptacles **19** extend transverse to a plate plane **25** spanned by cleat plate **3** end-to-end through the material of cleat plate **3**. A depth **27** of a receptacle **19**, which is measured transverse to respective longitudinal side **17** and parallel to plate plane **25**, is preferably about one third to two thirds, particularly preferably one third to one half, of a length **29** of end link **21**. A width **31** of receptacle **19** is preferably dimensioned such that it is slightly greater than a thickness **33** of end link **29**.

(26) Each cleat plate **3** preferably has a plate-shaped main body **35** and cleats **37**. A main body **35** without cleats **37** is illustrated in FIG. **5**. Main body **35** is preferably flat and spans plate plane **25** of a cleat plate **3**. Main body **35** is preferably formed from metal, in particular from sheet metal. Main body **35** substantially determines the shape of cleat plate **3**, i.e. the square shape in the example shown.

(27) Thickness **39** of main body **35** is preferably at least 15 mm, particularly preferably 20 to 30 mm. Main body **35** is preferably provided with a hole-shaped recess **41** at its center. In this way, firstly, material can be saved during production, which reduces the costs for a cleat plate **3**. Secondly, the weight of a cleat plate **3** is reduced by recess **41** so that a utility vehicle tire chain **1** equipped with cleat plates **3** can have less overall weight. As a result, the mounting on a utility vehicle tire can be simplified and the fuel consumption of the utility vehicle can be reduced.

(28) Recess **41** extends transverse to plate plane **25** through main body **35**. A diameter **43** of recess **41** is preferably smaller than half of a side length **45** of main body **35**. Diameter **43** is preferably greater than a third of side length **45**.

(29) Main body **35** and cleat plate **3**, respectively, have an underside **47**. Cleat plate **3** can rest with underside **47** on the tread surface of a utility vehicle tire (not shown). The underside **47** is preferably flat. It therefore forms base area **48** of cleat plate **3** or of main body **35**, respectively. An upper side **49** is disposed across from underside **47** transverse to plate plane **25**. In a mounted state

of a utility vehicle tire chain **1** on a utility vehicle tire, upper side **49** faces away from the tread surface of the utility vehicle tire.

(30) Cleats **37** extend away from upper side **49**. In particular, cleats **37** extend transverse to plate plane **25** away from upper side **49** of cleat plate **3**. Cleats **37** are preferably welded to main body **35**. In particular, they can be welded to upper side **49** of main body **35**.

(31) As an alternative thereto, cleats **37** can also penetrate somewhat into main body **35** or protrude therethrough. They can then be welded to main body **35** on the upper side and/or lower side **47**. As an alternative to the welded connection, cleats **37** can also be connected to main body **35** in some other way. For example, cleats **37** can be connected to main body **35** by way of a screw connection.

(32) The first embodiment of utility vehicle tire chain **1** is provided with cleat plates **3**, each of which comprises four cleats **37**. When viewed transverse to plate plane **25**, cleats **37** have a rectangular cross-section.

(33) Each cleat **37** is arranged close to a longitudinal side **17** of main body **35**. Each cleat **37** is arranged between respective longitudinal side **17** and recess **41**. Longitudinal sides **51** of cleats **37** run parallel to longitudinal sides **17** which are closest to respective cleats **37**. Each cleat **37** is arranged centrally between two receptacles **19** of respective longitudinal side **17**. A longitudinal side **51** of a cleat **37** can there end flush with the closed ends of receptacle **19**. This can be seen, for example, in FIGS. **2** and **4**. However, such an arrangement is not mandatory. Cleats **37** can also be arranged closer toward receptacle **41** or closer toward sides **17**. The configuration parallel to sides **17** is also not absolutely necessary. Cleats **37** can also extend obliquely relative to sides **17**.

(34) Cleats **37** are preferably manufactured from the same material as main body **35**. Cleats **37** can therefore have a cleat thickness **53** which corresponds to plate thickness **39** of main body **35**. A cleat height **55**, which is measured from upper side **49** to the end of cleat **37** disposed opposite upper side **49**, is preferably at least twice thickness **39** of main body **35**. Cleat height **55** is preferably at least 40 mm, preferably 50 to 70 mm.

(35) At its end disposed opposite upper side **49**, the cleat can be provided with a tip **57**. Tip **57** can further improve the traction of utility vehicle tire chain **1**. Tip **57** preferably has a length **59** which corresponds to a quarter to half of a cleat length **61**. Cleat length **61** is measured parallel to longitudinal sides **51**.

(36) The size of a cleat plate **3**, in particular the two-dimensional extension parallel to plate plane **25**, is preferably smaller than a tire tread area **63** of a utility vehicle tire for which utility vehicle tire chain **1** is intended. When manufacturing a utility vehicle tire chain **1**, the size of the utility vehicle tire for which utility vehicle tire chain **1** is intended is generally known. Consequently, the dimensions of utility vehicle tire chain **1**, in particular of cleat plates **3**, can be adapted. A tire tread area **63** is indicated by the dashed line in FIG. **1**. For the reason that cleat plates **3** are smaller than tire tread area **63**, excessive bending of cleat plates **3** can be prevented when the tire rolls.

(37) A second embodiment of a utility vehicle tire chain **1** according to the invention shall be described hereafter with reference to FIGS. **7** and **8**. FIG. **7** shows utility vehicle tire chain **1** in an unmounted state and FIG. **8** in a mounted state on a utility vehicle tire **65**, wherein side strands **10** of utility vehicle tire chain **1** are not shown in FIG. **8** for the sake of clarity.

(38) In the following, only the differences between utility vehicle tire chain **1** and the first embodiment described with reference to FIGS. **1** to **6** shall be discussed.

(39) Cleat plates **3** of the second embodiment are connected to one another each with only one chain strand **5**. In other words, each cleat plate **3** is connected via a respective transverse connection **9** to a cleat plate **3** that is adjacent transverse to circumferential direction **U**. In circumferential direction **U**, each cleat plate **3** is connected to a preceding and a subsequent cleat plate **3** via a respective longitudinal connection **7**.

(40) Each cleat plate **3** is connected to two side strands **10**, wherein one side strand **10** each of a cleat plate **3** is connected via a ring **67** to a side strand **10** of a cleat plate **3** that is adjacent in circumferential direction **U**. Due to the above-described arrangement of chain strands **5**, each cleat

plate **3** comprises two receptacles **19** on one side for connection to side strands **10** and one receptacle **19** each on the other three sides for chain strands **5** for the direct connection to further cleat plates **3**

(41) Each cleat plate **3** is provided with two arrow-shaped cleats **37**, wherein two arrow-shaped cleats **37** each point in the same direction. Each arrow-shaped cleat **37** is formed from two rectangular blocks **69** arranged at a right angle to one another. The arrow-shaped cleats each point in the direction of the closest side strands **10**, so that cleats **37** of two cleat plates **3** arranged at the same height in circumferential direction **U** point in the opposite direction and away from one another. However, such an arrangement is not mandatory. Cleats **37** can also have other shapes than the arrow shape described above. For example, they can also have a zigzag shape, wherein such a zigzag shape can be obtained by using three blocks **69** to form a cleat **37**.

(42) Cleat plates **3** of the second embodiment can be formed without a recess **41**. However, as an alternative thereto, at least recess **41** can also be present in a cleat plate **3**.

(43) In a state of the utility vehicle tire chain being mounted on utility vehicle tire **65**, circumferential direction **U** of the tire chain can coincide with a circumferential direction of utility vehicle tire **65**. Circumferential direction **U** then runs around an axis **71** of utility vehicle tire **65**.

(44) A third embodiment of a utility vehicle tire chain **1** according to the invention shall be described hereafter with reference to FIGS. **9** and **11**. FIG. **9** shows a schematic perspective view of utility vehicle tire chain **1** in an unmounted state, FIG. **10** shows a perspective view of a main body **35**, and FIG. **11** shows a perspective schematic illustration of a cleat of utility vehicle tire chain **1** from FIG. **9**.

(45) The basic structure of utility vehicle tire chain **1** consisting of cleat plates **3** and chain strands **5** corresponds substantially to the structure of the first embodiment described with reference to FIGS. **1** to **6**.

(46) For the sake of brevity, only the differences to the first embodiment of utility vehicle tire chain **1** according to the invention and its components shall be discussed below.

(47) Cleat plates **3** of the third embodiment are each provided with three cleats **37**. Cleats **37** are arranged substantially like three of four cleats **37** of the first embodiment. This means that two cleats **37** are arranged opposite one another in circumferential direction **U** and a third cleat **37** is disposed along circumferential direction **U** between the two first-mentioned cleats **37**. Overall, three cleats **37** form a U-shaped arrangement, wherein three cleats **37** are spaced from one another. A recess **41** extends into or through main body **35** between three cleats **37**.

(48) Disposed opposite cleat **37**, which is not disposed opposite second cleat **37** beyond recess **41**, is a lateral indentation **73** in main body **35**, which likewise represents a recess **41**. In other words, lateral indentation **73** is located in a region between the ends of the legs of an imaginary U-shaped structure that is spanned by three cleats **37**.

(49) Lateral indentation **73** extends along plate plane **25** into main body **35** and is formed end-to-end transverse to plate plane **25**. A web **75** made of the material of main body **35** extends between lateral indentation **73** and recess **41** arranged at the center of main body **35**. Indentation **73** has an overall V-shape, wherein tip **74** of the V-shape points into main body **35** in the direction of centrally arranged recess **41**.

(50) Aligned within utility vehicle tire chain **1** are two juxtaposed cleat plates **3** in such a way that their lateral indentations **73** are disposed opposite one another transverse to circumferential direction **U**, wherein their open sides **76** are aligned toward each other. In other words, tips **74** of the V-shapes point away from one another. Indentations **73** enclose a free area **77** between them which is defined transverse to circumferential direction **U** by main bodies **35** and in circumferential direction **U** by chain strands **5** which form transverse connections **9**. Free areas **77** can improve the self-cleaning of utility vehicle tire chain **1** because fouling such as mud or stones can only stick poorly in this region.

(51) In circumferential direction **U**, two successive cleat plates **3** are connected to only one chain

strand 5 each. Consequently, each cleat plate 3 on its longitudinal side 17 disposed in circumferential direction U comprises a receptacle 19 for an end link 21 of a chain strand 5. Cleats 37 arranged on these sides 17 are arranged such that they bridge receptacles 19. Cleats 37 run with their longitudinal sides 51 transverse to the direction in which receptacles 19 extend into the material of main body 35. This direction is the same direction in which depth 27 of receptacle 19 extends. Cleat thickness 53 can there substantially correspond to depth 27 of receptacle 19. However, this is not mandatory. Cleat thickness 53 can also be greater or smaller than depth 27 of receptacle 19. Cleats 37 bridging receptacles 19 can be aligned with those receptacles 19 which are open in the direction of cleat plate 3 that is disposed oppositely, transverse to circumferential direction U.

(52) Receptacles 19, which are open in the direction of cleat plate 3 that is disposed oppositely, transverse to circumferential direction U, can be offset along circumferential direction U relative to receptacles 19 which are used to connect to side strands 10. In other words, receptacles 19, which open toward cleat plate 3 which is disposed oppositely, transverse to circumferential direction U, can have a greater spacing 79 from one another than two oppositely disposed receptacles 19 which serve to connect side strands 10. Last-mentioned receptacles 19 have spacing 81.

(53) At least cleats 37, which are arranged such that they bridge a receptacle, can on their undersides 83 respectively comprise a recess 85 extending into cleat 37. Recess 85 has a length 87 which preferably corresponds to at least width 31 of a receptacle 19. For the sake of simplicity, length 87 of the recess can be equal to length 59 of the tip of cleat 37. In the fully mounted state, an end link 21 of a chain strand 5 can protrude into recess 85.

(54) A height 89 of recess 85 is there preferably selected such that an end link 21 arranged in receptacle 19 does not hit cleat 37. Overall, a compact structure of a cleat plate 3 is possible due to cleats 37 each bridging receptacles 19. In addition, cleats 37 each form a protection for receptacle 19 which they bridge. Likewise, cleats 37 can protect an end link 21 received in receptacle 19 or another part of a chain strand 5 received in receptacle 19.

REFERENCE NUMERALS

(55) 1 utility vehicle tire chain 3 cleat plate 5 chain strand 7 longitudinal connection 9 transverse connection 10 side strand 11 row of cleat plates 13 continuous chain mesh plane 15 plane of symmetry 17 longitudinal side 19 receptacle for end link 21 end link 23 welded connection 25 plate plane 27 depth of receptacle 29 end link 31 width of the receptacle 33 thickness of the end link 35 main body 37 cleat 39 thickness of the body 41 recess 43 diameter 45 side length of the main body 47 underside 48 base area 49 upper side 51 longitudinal sides of the cleat 53 cleat thickness 55 cleat height 57 tip 59 length of tip 61 cleat length 63 tire tread area 65 utility vehicle tire 67 ring 69 block 71 axis of the utility vehicle tire 73 lateral indentation 74 tip 75 web 76 open side 77 free area 79 spacing 81 spacing 83 underside 85 recess 87 length of the recess 89 height of the recess U circumferential direction

Claims

1. A utility vehicle tire chain, which comprises a plurality of cleat plates directly or indirectly connected to one another by chain strands, wherein each cleat plate comprises at least one cleat to improve traction which extends away from an upper side of said cleat plate, wherein at least one of said cleat plates has a plate-shaped main body with at least one recess extending transverse to a plate plane into said main body.
2. The utility vehicle tire chain according to claim 1, wherein at least one cleat is welded to a plate-shaped main body of said cleat plate.
3. The utility vehicle tire chain according to claim 1, wherein at least one cleat is monolithically formed with a plate-shaped main body of said cleat plate.
4. The utility vehicle tire chain according to claim 1, wherein at least one of said cleat plates has a

rectangular main body.

5. The utility vehicle tire chain according to claim 1, wherein at least one of said cleat plates has a plate-shaped main body with at least two receptacles, in each of which an end link of a chain strand is received.
 6. The utility vehicle tire chain according to claim 1, wherein a cleat height corresponds to at least twice the thickness of a plate-shaped main body of said cleat plate.
 7. The utility vehicle tire chain according to claim 1, wherein at least one of said cleat plates has a plate-shaped main body, the thickness of which is at least 15 mm.
 8. The utility vehicle tire chain according to claim 1, wherein said utility vehicle tire chain comprises at least one, row of cleat plates extending along a circumferential direction of said utility vehicle tire chain.
 9. The utility vehicle tire chain according to claim 8, wherein said utility vehicle tire chain comprises at least two rows of cleat plates and wherein at least two cleat plates are arranged adjacent to one another transverse to said circumferential direction.
 10. The utility vehicle tire chain according to claim 8, wherein said utility vehicle tire chain comprises at least two rows of cleat plates and wherein at least two cleat plates are arranged offset from one another in said circumferential direction.
 11. The utility vehicle tire chain according to claim 1, wherein said utility vehicle tire chain is symmetrical with respect to a plane of symmetry extending transverse to a tire axis.
 12. The utility vehicle tire chain according to claim 1, wherein at least one cleat plate is smaller than a tire tread area of a utility vehicle tire.
 13. A utility vehicle tire with a utility vehicle tire chain according to claim 1.
 14. A cleat plate for a utility vehicle tire chain according to claim 1.
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